Rui Pan

ruipan.xyz | github.com/ruipeterpan | ruipan@princeton.edu | Google Scholar

Education

Princeton University — Princeton, NJ, USA
Ph.D. in Computer Science — *Aug 2022 – Sep/Oct 2026 (Expected)*
- M.A. in Computer Science, May 2025 (conferred en route to Ph.D.).
- Advisor: Prof. Ravi Netravali.

University of Wisconsin–Madison — Madison, WI, USA
B.S. in Computer Science and B.S. in Mathematics — *Sep 2018 – Dec 2021*
- GPA: 3.96/4.00.
- Advisor: Prof. Shivaram Venkataraman.

Research Interests

My research interests lie in **systems and algorithms for efficient LLMs across the stack**. My PhD work centers on model-system co-design for efficient LLM inference — studying how inference optimizations such as prefix caching and speculative decoding interact with emerging model architectures, including hybrid, reasoning, and diffusion LLMs. More broadly, I am interested in LLM systems and infrastructure spanning the full stack, from inference to pre-/post-training. My research has been adopted by open-source LLM serving systems such as vLLM and SGLang, and has been recognized with an MLSys Outstanding Paper Award Honorable Mention, a Jane Street Graduate Research Fellowship Finalist Award, an MLCommons ML and Systems Rising Stars Award, and research grants from Google, a16z, Modal, and Lambda.

First-Author Publications (* denotes equal contribution)

[5] Fail Fast, Win Big: Rethinking the Drafting Strategy in Speculative Decoding via Diffusion LLMs. **Rui Pan**, Zhuofu Chen, Hongyi Liu, Arvind Krishnamurthy, Ravi Netravali. In *submission*; arXiv preprint available (2025).

[4] SpecReason: Fast and Accurate Inference-Time Compute via Speculative Reasoning. **Rui Pan**, Yinwei Dai, Zhihao Zhang, Gabriele Oliaro, Zhihao Jia, Ravi Netravali. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems* **(NeurIPS '25)**. **Spotlight Paper** at *The NeurIPS 2025 Workshop on Efficient Reasoning* **(ER '25)**.

[3] Marconi: Prefix Caching for the Era of Hybrid LLMs. **Rui Pan**, Zhuang Wang, Zhen Jia, Can Karakus, Luca Zancato, Tri Dao, Yida Wang, Ravi Netravali. In *The Eighth Annual Conference on Machine Learning and Systems* **(MLSys '25)**. **Outstanding Paper Award, Honorable Mention.** **Integrated into vLLM and SGLang!**

[2] Apparate: Rethinking Early Exits to Tame Latency-Throughput Tensions in ML Serving. Yinwei Dai*, **Rui Pan***, Anand Iyer, Kai Li, Ravi Netravali. In *The 30th Symposium on Operating Systems Principles* **(SOSP '24)**.

[1] Efficient Flow Scheduling in Distributed Deep Learning Training with Echelon Formation. **Rui Pan***, Yiming Lei*, Jialong Li, Zhiqiang Xie, Binhang Yuan, Yiting Xia. In *The 21st ACM Workshop on Hot Topics in Networks* **(HotNets '22)**.

Other Publications

[13] Slipstream: Trajectory-Grounded Compaction Validation for Long-Horizon Agents. Zhuofu Chen, **Rui Pan**, Yinwei Dai, Ravi Netravali. In *submission*; arXiv preprint available (2026).

[12] Detection-Guided Attention Steering for Vision Language Models. Alan Zhang, **Rui Pan**, Mike Wong, Ravi Netravali. In *The ICML 2026 Workshop on Multimodal AI Agents* **(SCALE @ ICML '26)**. Also in *The ICML 2026 Workshop on Efficient Multimodal Question Answering* **(EMM-QA @ ICML '26)**.

[11] A benchmark of expert-level academic questions to assess AI capabilities. Center for AI Safety, Scale AI, HLE Contributors Consortium. In *Nature* **(Nature '26)**.

[10] Optimizing Mixture-of-Experts Inference Time via Model Deployment and Communication Scheduling. Jialong Li, Shreyansh Tripathi, Lakshay Rastogi, Yiming Lei, **Rui Pan**, Yiting Xia. In *IEEE/ACM Transactions on Networking* **(ToN '26)**.

[9] METIS: Fast Quality-Aware RAG Systems with Configuration Adaptation. Siddhant Ray, **Rui Pan**, Zhuohan Gu, Kuntai Du, Shaoting Feng, Ganesh Ananthanarayanan, Ravi Netravali, Junchen Jiang. In *The 31st Symposium on Operating Systems Principles* **(SOSP '25)**.

[8] Mowgli: Passively-Learned Rate Control for Real-Time Video. Neil Agarwal, **Rui Pan**, Francis Y. Yan, Ravi Netravali. In *The 22nd USENIX Symposium on Networked Systems Design and Implementation* **(NSDI '25)**.

[7] Improving DNN Inference Throughput Using Practical, Per-Input Compute Adaptation. Anand Iyer, Mingyu Guan, Yinwei Dai, **Rui Pan**, Swapnil Ghandi, Ravi Netravali. In *The 30th Symposium on Operating Systems Principles* **(SOSP '24)**.

[6] Shockwave: Fair and Efficient Cluster Scheduling for Dynamic Adaptation in Machine Learning. Pengfei Zheng, **Rui Pan**, Tarannum Khan, Shivaram Venkataraman, Aditya Akella. In *The 20th USENIX Symposium on Networked Systems Design and Implementation* **(NSDI '23)**.

Experience

Google, SystemsResearch@Google — Sunnyvale, CA, USA
Student Researcher — *Jun 2025 – Dec 2025*
- Mentor: Prof. Arvind Krishnamurthy.
- Topic: Systems for efficient LLM inference.
- Developed and open-sourced FailFast, a speculative decoding framework that leverages Diffusion Large Language Models (dLLMs) as drafters to achieve lossless acceleration of autoregressive LLMs.
- Achieved up to 5× speedup over vanilla decoding and 2× speedup over state-of-the-art baselines such as EAGLE-3 across diverse benchmarks.

AWS AI, Machine Learning Systems Team — Santa Clara, CA, USA
Applied Scientist Intern — *May 2024 – Dec 2024*
- Mentors: Dr. Zhen Jia, Dr. Zhuang Wang, Dr. Can Karakus, and Dr. Yida Wang.
- Topic: Systems for efficient LLM inference.
- Researched efficient prefix caching for Hybrid LLMs (e.g., Qwen3.5, Gemini) that combine Attention layers with subquadratic layers (e.g., Mamba, Gated DeltaNet, Sliding Window Attention).
- Implemented and open-sourced Marconi, a radix-tree-based prefix management system partially integrated into vLLM and SGLang, popular LLM inference frameworks with 80k+ and 20k+ GitHub stars.
- Improved efficiency by up to 34.4× compared to state-of-the-art prefix caching systems such as vLLM and SGLang.
- Received the Outstanding Paper Award (Honorable Mention) at MLSys 2025.

Princeton University — Princeton, NJ, USA
Graduate Research Assistant — *Aug 2022 – Present*
- Advisor: Prof. Ravi Netravali.
- Topic: Systems for efficient ML/LLM inference.
- SpecReason: Low-latency reasoning model inference paradigm that generalizes speculative decoding by relaxing per-token equivalence into semantic equivalence, achieving complementary efficiency gains.
- METIS: Low-latency RAG system that jointly adapts RAG configurations per query and schedules them.
- Apparate and E3: Low-latency ML and LLM inference via early exiting.
- Mowgli: Offline reinforcement learning for rate control adaptation in video conferencing.
- Ongoing projects: Efficient multi-round agents, efficient multimodal model inference, and efficient deep research agents.

Max Planck Institute for Informatics, Network and Cloud Systems Group — Saarbrücken, Germany
Research Intern — *Feb 2022 – Aug 2022*
- Advisor: Prof. Yiting Xia.
- Topic: Networked systems for efficient ML and LLM training.
- Researched network traffic patterns of parallelization strategies in distributed deep learning training. Proposed a network abstraction for flow scheduling in large-scale ML/LLM training clusters.

Professional Activities

Conference Reviewer: MLSys 2026, ICML 2026, NeurIPS 2026.

Workshop Reviewer: NeurIPS 2025 Workshop on Efficient Reasoning; KDD 2025 Workshop on Inference Optimization for Generative AI; ICLR 2026 Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning; COLM 2026 Workshop on Efficient Reasoning.

Journal Reviewer: Transactions on Machine Learning Research; IEEE Transactions on Parallel and Distributed Systems; IEEE Transactions on Networking; IEEE Transactions on Cloud Computing; IEEE Transactions on Mobile Computing; IEEE Transactions on Computers; IEEE Transactions on Machine Learning in Communications and Networking; IEEE Communications Letters; Computer Networks; Future Generation Computer Systems.

Artifact Evaluation Committee: MLSys '23, OSDI '23, ATC '23.

Student Volunteer: CMMRS '22, N2Women@SIGCOMM '22.

Teaching Assistant: COS 316 Principles of Computer System Design (Fall 2023); COS 598D Systems and Machine Learning (Spring 2024).

Invited Talks

- Distributed Systems Lab, University of Pennsylvania, Dec 2024.
- IEEE EDS/EPS Lunch Series: Frontier AI, UCLA, Mar 2025.
- ML Systems Group, UCSD, May 2025.
- KDD 2025 Workshop on Inference Optimization for Generative AI, Aug 2025.
- Jane Street Graduate Research Fellowship Workshop, Apr 2026.

Awards & Honors

- MLSys Outstanding Paper Award, Honorable Mention, 2025.
- MLSys Travel Grant, 2025 ($1,000).
- NeurIPS Workshop on Efficient Reasoning, Spotlight Paper, 2025.
- Jane Street Graduate Research Fellowship, Finalist Award, 2026 ($5,000).
- Lambda Research Grant, 2026 ($2,000).
- Google Cloud Research Credits, 2026 ($1,000).
- MLCommons ML and Systems Rising Stars Award, 2026.
- a16z GPU Compute Grant, 2026 ($10,000).
- Modal Academic Compute Grant, 2026 ($10,000).
- ICML Gold Reviewer, 2026.

Last updated: Jun 2026